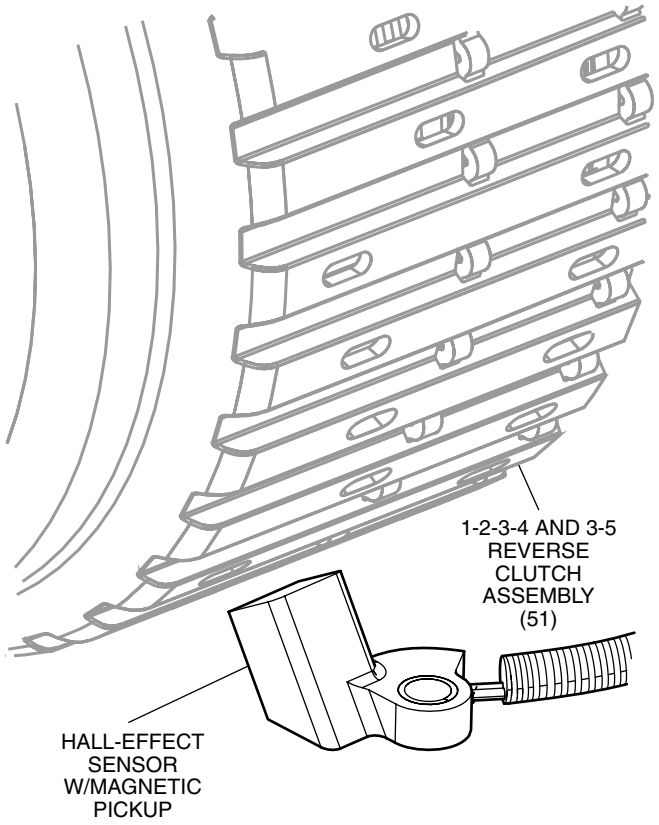
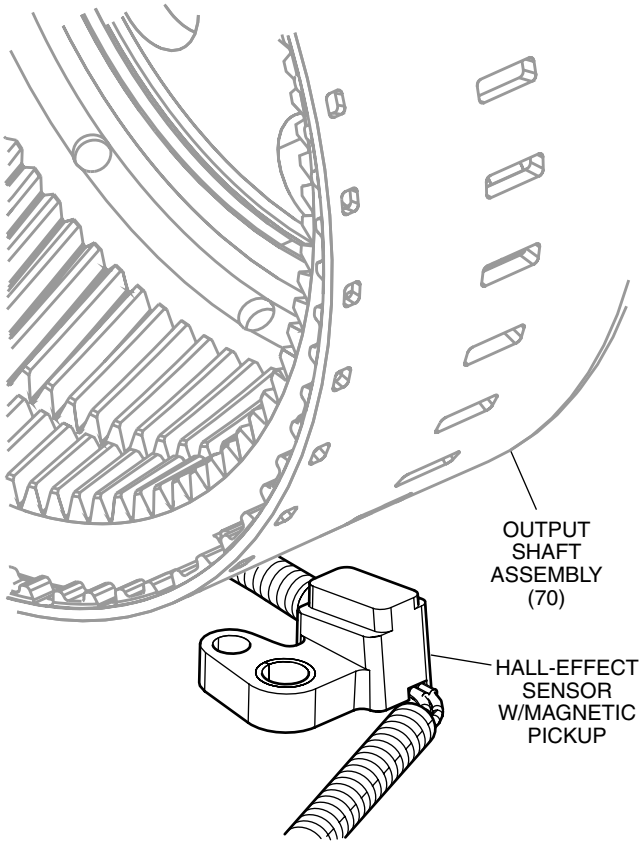


INPUT SPEED SENSOR

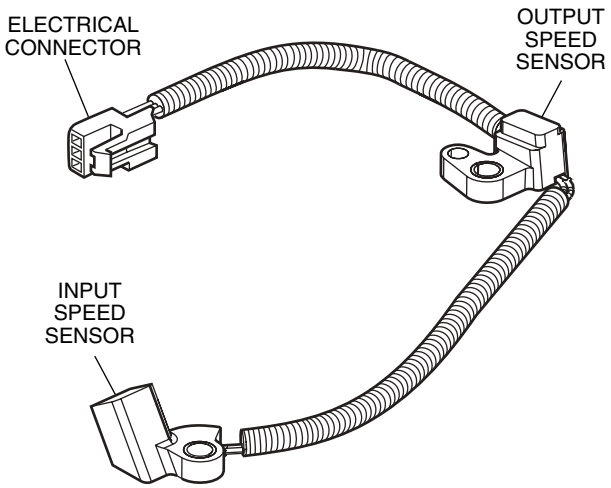


OUTPUT SPEED SENSOR

TRANSMISSION SPEED SENSORS

Automatic Transmission Input (Shaft) Speed (A/T ISS) Sensor:

The input speed sensor (ISS) is a hall-effect type sensor. The ISS mounts to the control valve upper body assembly and connects to the control solenoid (w/body and TCM) valve assembly through a wire harness and connector. The sensor faces the 1-2-3-4 and 3-5-R clutch housing and is triggered by splines on the housing outside diameter. The sensor receives 8.3–9.3 volts on the ISS/OSS supply voltage circuit from the TCM. As the 1-2-3-4 and 3-5-R clutch housing rotates, the sensor produces a signal frequency based on the spline profile and speed of the 1-2-3-4 and 3-5-R clutch housing. This signal is transmitted through the ISS signal circuit to the TCM. The TCM uses the ISS signal to determine line pressure, transmission shift timing, torque converter clutch (TCC) slip speed and gear ratio.



If the TCM detects an improper signal from the transmission input or output speed sensors, a DTC will be activated.

Automatic Transmission Output Speed (A/T OSS) Sensor:

The output speed sensor (OSS) is also a hall-effect type sensor. The OSS mounts to the control valve upper body assembly and connects to the control solenoid (w/body and TCM) valve assembly through a wire harness and connector. The sensor faces the output shaft and is triggered by slots cut into the housing. The sensor receives 8.3–9.3 volts on the ISS/OSS supply voltage circuit from the TCM. As the output shaft rotates, the sensor produces a signal frequency based on the machined slots and speed of the output shaft. This signal is transmitted through the OSS signal circuit to the TCM. The TCM uses the OSS signal to determine line pressure, transmission shift timing, vehicle speed and gear ratio.

Figure 37